

Statistical mechanics of decentralized agentic coordination: equilibrium limits, entropy production, and finite-size structure on multiplex networks

Anonymous working draft

V9 formulation study, August 17, 2026

Abstract

We formulate a decentralized autonomous-agent system as a coupled belief–action process on a multiplex network. Each agent has private state, persistent memory, an inbox and outbox, and a local stochastic policy; a scheduler supplies update opportunities but never chooses collective actions. In a restricted static, symmetric, memoryless limit, the local heat-bath rule obeys detailed balance with respect to an explicit two-layer Hamiltonian. Exact enumeration validates the Gibbs stationary distribution and the nonequilibrium free-energy identity. Directed communication, private-field disorder, memory, task injection, and topology changes break this equilibrium structure. We quantify those departures using probability currents and entropy production, and numerically examine order parameters, Binder cumulants, response peaks, relaxation, hysteresis, topology, and finite-size effects. The numerical evidence is described as a finite-size crossover, not assumed to be a thermodynamic-limit phase transition. Generalized Tsallis entropies are evaluated prospectively but did not establish independent value beyond Shannon entropy. Humanitarian logistics and defensive cyber-utility restoration are included only as interpretive mappings of the same microdynamics.

1 Introduction

Autonomous multi-agent systems are often described in the language of coordination, consensus, congestion, and cascades. Statistical mechanics offers a more disciplined question: can one specify a microstate and stochastic local dynamics for such a system, identify an equilibrium reference measure, and then measure how operational features drive it away from that reference? This is different from attaching an “energy” label to an arbitrary performance score.

Kinetic Ising dynamics supplies a canonical local stochastic update whose reversible limit is analytically controlled [1]. Logit revision also has a direct interpretation in local strategic interaction [2]. Network substrates change collective critical behavior [3], and multiplex networks provide a natural representation when communication and operational

dependence are not the same graph [4]. These ideas have been used broadly in agent-based and social dynamics [5], but a distinction is often lost between agents as independent decision-making units and spins as centrally updated labels.

We make that distinction executable. An agent owns its belief, action, memory, workload, private fields, commitments, inbox, and outbox. It samples a local policy from authorized information. The simulator schedules updates and applies transitions, but receives no authority to replace the proposal by a centralized choice. The primary model is deliberately minimal so its equilibrium limit can be proved. Application mappings are secondary.

The contributions are: (i) an explicit coupled belief–action Hamiltonian and detailed-balance proof; (ii) an exact finite-state validation including the Gibbs law and a relative-entropy free-energy identity; (iii) a controlled route to nonequilibrium dynamics and exact stationary entropy production under directed coupling; (iv) numerical finite-size, topology, disorder, relaxation, and hysteresis analyses; and (v) a precise account of where Shannon entropy is or is not supplemented by generalized entropy.

2 Autonomous-agent microdynamics

2.1 Microstate and multiplex structure

There are N persistent agents. Agent i has a binary belief or intention $b_i \in \{-1, +1\}$, an executed or committed action $a_i \in \{-1, +1\}$, a slow memory m_i , and a workload w_i . The full equilibrium microstate is

$$x = (\mathbf{b}, \mathbf{a}) \in \{-1, +1\}^{2N}. \quad (1)$$

Memory and workload are held fixed at zero in the equilibrium reference and are activated only in driven extensions. Communication and operational dependence are represented by adjacency matrices $A^{(c)}$ and $A^{(d)}$, respectively. They can differ in topology and, outside equilibrium, in direction and time.

Figure 1 shows the executable information boundary. A policy call receives local private state and explicitly delivered messages, never the global microstate or another private vault.

2.2 Equilibrium-reference Hamiltonian

All quantities below are nondimensional. We define

$$\begin{aligned} H(x) = & -\frac{J_b}{2} \sum_{ij} A_{ij}^{(c)} b_i b_j - \frac{J_a}{2} \sum_{ij} A_{ij}^{(d)} a_i a_j - K \sum_i a_i b_i \\ & - \sum_i h_i b_i - \sum_i g_i a_i. \end{aligned} \quad (2)$$

Here J_b controls belief alignment, J_a action compatibility, K local belief–action consistency, h_i private evidence, and g_i local task drive. This is a genuine Hamiltonian only when both

Multiplex belief-action microdynamics

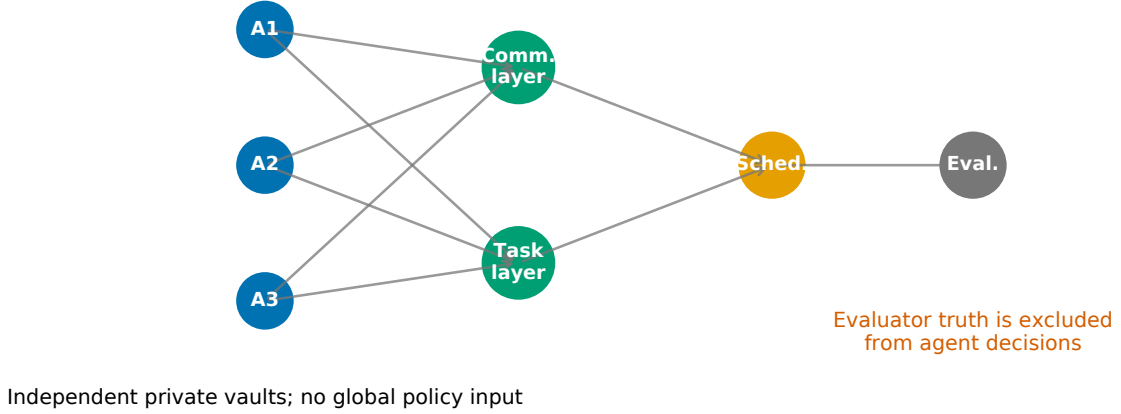


Figure 1: Independent-agent and multiplex formulation. The evaluator can reconstruct the global microstate for analysis, but that state is not an input to any deployable local policy.

layers are symmetric and static, all agents share a temperature, and memory, delay, loss, and task injection are disabled. Elsewhere we call its symmetric part an energy diagnostic, not an equilibrium energy.

2.3 Local heat-bath decisions

One agent-variable is selected uniformly at each asynchronous update. Conditional on the rest of the microstate, the local fields are

$$\ell_i^{(b)} = J_b \sum_j A_{ij}^{(c)} b_j + K a_i + h_i, \quad (3)$$

$$\ell_i^{(a)} = J_a \sum_j A_{ij}^{(d)} a_j + K b_i + g_i. \quad (4)$$

The selected variable s_i is set to +1 with

$$\Pr(s_i = +1 \mid \mathcal{F}_i) = [1 + \exp(-2\ell_i/T)]^{-1}. \quad (5)$$

The decision temperature $T > 0$ is therefore the stochasticity of a local softmax policy. It is specified before simulation and is not fitted from performance.

For large equilibrium runs we use chromatic block Gibbs sampling. Nodes of a color class are an independent set, so their conditional updates factorize. Each block kernel preserves the same Gibbs measure; color order is randomized between sweeps. The exact detailed-balance calculation uses the asynchronous single-variable kernel, not the block implementation.

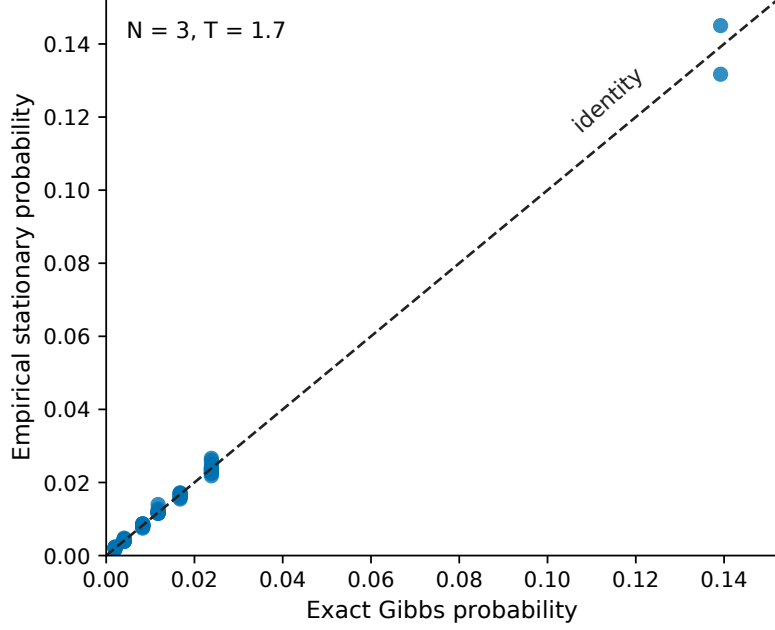


Figure 2: Empirical and exact Gibbs probabilities for every microstate at the prespecified representative temperature. Samples are generated by local asynchronous agent updates.

3 Equilibrium reference

3.1 Detailed balance and stationary distribution

Suppose $A^{(c)}$ and $A^{(d)}$ are symmetric and static, $m_i = w_i = 0$, and agents share T . Let x and y differ in exactly one variable. From equation (5),

$$\frac{W_{xy}}{W_{yx}} = \exp \left[-\frac{H(y) - H(x)}{T} \right]. \quad (6)$$

For

$$\pi(x) = Z^{-1} \exp[-H(x)/T], \quad Z = \sum_x \exp[-H(x)/T], \quad (7)$$

equation (6) implies $\pi(x)W_{xy} = \pi(y)W_{yx}$. Self transitions are trivially balanced, and non-neighboring state pairs have zero transition probability in both directions. Thus the finite irreducible heat-bath chain is reversible and equation (7) is its unique stationary distribution.

Exact enumeration at $N = 3$ gives a maximum detailed-balance flux residual of 6.51×10^{-18} . The maximum empirical total-variation distance from the Gibbs law over the frozen temperature grid is 0.0219. A representative comparison appears in figure 2.

3.2 Nonequilibrium free-energy identity

For any distribution p over equilibrium microstates, define

$$F[p] = \sum_x p(x)H(x) + T \sum_x p(x) \log p(x). \quad (8)$$

Substitution of $\log \pi(x) = -H(x)/T - \log Z$ gives the exact identity

$$F[p] - F[\pi] = TD_{\text{KL}}(p\|\pi) \geq 0. \quad (9)$$

The largest numerical closure residual in the exact suite is 1.86×10^{-15} . This identity requires the explicit Hamiltonian and reference law; it is not asserted for the directed or driven models.

4 Mean-field approximation and stability

For homogeneous mean degrees z_c, z_d and uniform fields, the mean-field order parameters satisfy

$$m_b = \tanh[(J_b z_c m_b + K m_a + h)/T], \quad (10)$$

$$m_a = \tanh[(J_a z_d m_a + K m_b + g)/T]. \quad (11)$$

At zero field, linearization about $(m_b, m_a) = (0, 0)$ gives

$$\begin{pmatrix} m_b \\ m_a \end{pmatrix}_{t+1} \simeq \frac{1}{T} \begin{pmatrix} J_b z_c & K \\ K & J_a z_d \end{pmatrix} \begin{pmatrix} m_b \\ m_a \end{pmatrix}_t. \quad (12)$$

The disordered state loses stability when the largest eigenvalue equals T . For quenched independent communication dilution p_c , the first approximation is $z_c \mapsto p_c z_c$. Correlations, finite components, topology, and disorder make this a boundary prediction rather than an exact transition temperature.

5 Nonequilibrium dynamics and entropy production

Private heterogeneous fields preserve a Gibbs description when static, but directed interactions need not derive from equation (2). Delays, stale messages, loss, partitions, persistent memory, workload injection, and time-dependent fields further break the reversible reference.

For a finite stationary Markov chain with stationary probabilities P_x and kernel W_{xy} , define currents $J_{xy} = P_x W_{xy} - P_y W_{yx}$. The exact total entropy-production rate per update is [6, 7]

$$\dot{S}_{\text{tot}} = \frac{1}{2} \sum_{x,y} J_{xy} \log \frac{P_x W_{xy}}{P_y W_{yx}} \geq 0. \quad (13)$$

It is zero in the reversible suite to within 1.42×10^{-31} . Breaking communication reciprocity produces the response in figure 3; the high-asymmetry mean is 0.0531 (95% interval [0.0434, 0.0631], $n = 24$).

For time-dependent application protocols we log work from field changes, $\delta W = -\sum_i \delta h_i b_i - \sum_i \delta g_i a_i$, and transition affinities $\log(W_{xy}/W_{yx})$. We do not claim a Crooks relation or split housekeeping and excess terms because the application-level coarse state is not shown to satisfy the required trajectory assumptions [8, 9].

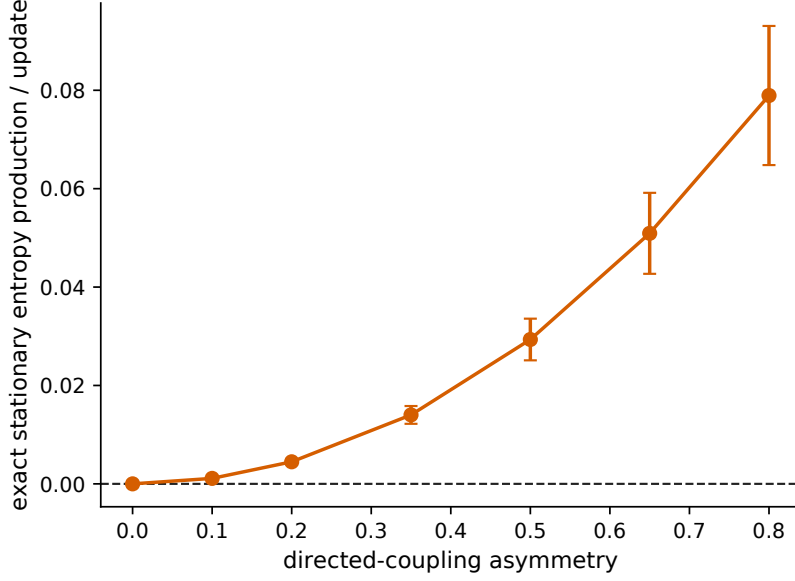


Figure 3: Exact stationary entropy production after directed communication breaks detailed balance. Intervals are across independent edge-orientation realizations, not time steps.

6 Observables and numerical method

The primary order parameters are belief order $m_b = N^{-1} \sum_i b_i$, action order m_a , combined order $m = (m_b + m_a)/2$, and belief–action consistency $q_{ba} = N^{-1} \sum_i a_i b_i$. Response quantities are

$$\chi = N \text{Var}(m)/T, \quad (14)$$

$$C = \text{Var}(H)/(NT^2), \quad (15)$$

$$U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}. \quad (16)$$

We also record activity, mutual information between belief and action, integrated autocorrelation time, effective sample size, and relaxation from an ordered state. Binder intersections and response peaks follow standard finite-size logic [10]; on network substrates, topology-dependent critical behavior is expected [3].

The frozen protocol contains five lattice sizes, six independent realizations per finite-size cell, four topology families, communication dilution, private field disorder, and fresh seeds. Burn-in and thinning were selected from a retained pilot before formal execution. Correlated Monte Carlo samples are not treated as inferential replicates.

Configurational Shannon entropy is computed from the empirical coarse magnetization distribution. The secondary family

$$S_q(p) = \frac{1 - \sum_k p_k^q}{q - 1} \quad (17)$$

uses a prespecified small set of q , including the Shannon limit and the $q = 2$ Gini–Simpson

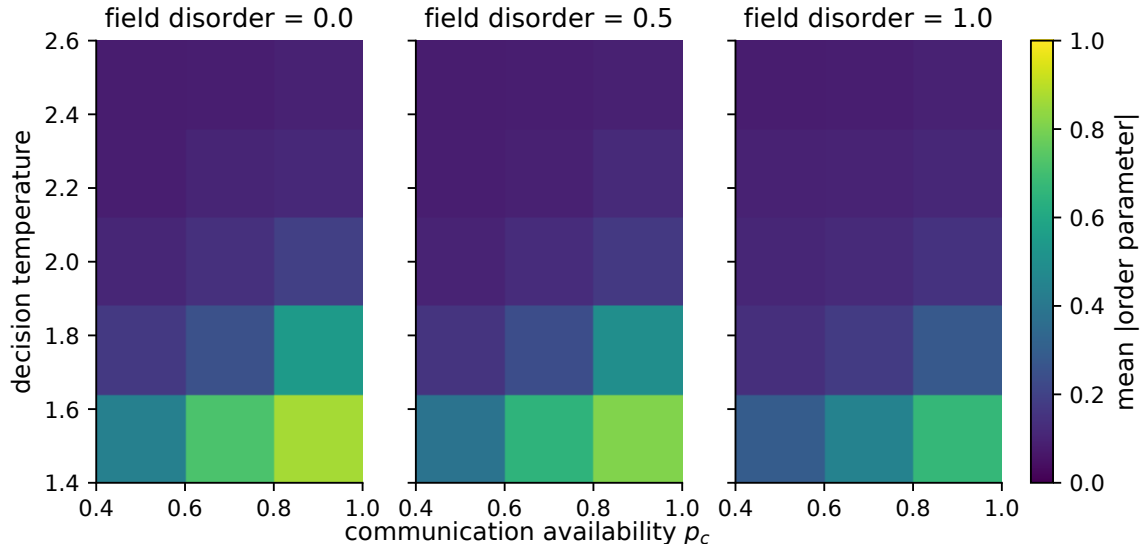


Figure 4: Numerical order surface at $N = 144$ on the regular multiplex. Each cell averages independent graph/seed realizations from the frozen grid.

form [11]. This comparison is diagnostic: no q is selected post hoc. Equal trajectory lengths make the estimators comparable within the frozen grid, but no separate finite-sample bias correction is claimed; that omission limits interpretation of small entropy differences.

7 Finite-size structure

Figure 4 maps order across decision temperature, communication availability, and private-field disorder. Reduced communication shifts and broadens the ordered-to-fragmented boundary in the direction predicted by the mean-field eigenvalue, while topology and disorder introduce finite-size heterogeneity.

The Binder curves in figure 5 yield a median first-intersection candidate of $T = 1.459$. Because the crossings are multiple and noncoherent for two of four adjacent size pairs, we use the phrase *finite-size crossover* rather than claim a demonstrated thermodynamic-limit phase transition. Susceptibility and heat-capacity analogues in figure 6 peak in the same broad region. The fitted finite-size exponent for susceptibility-peak growth is 0.251; it is descriptive rather than a universality estimate.

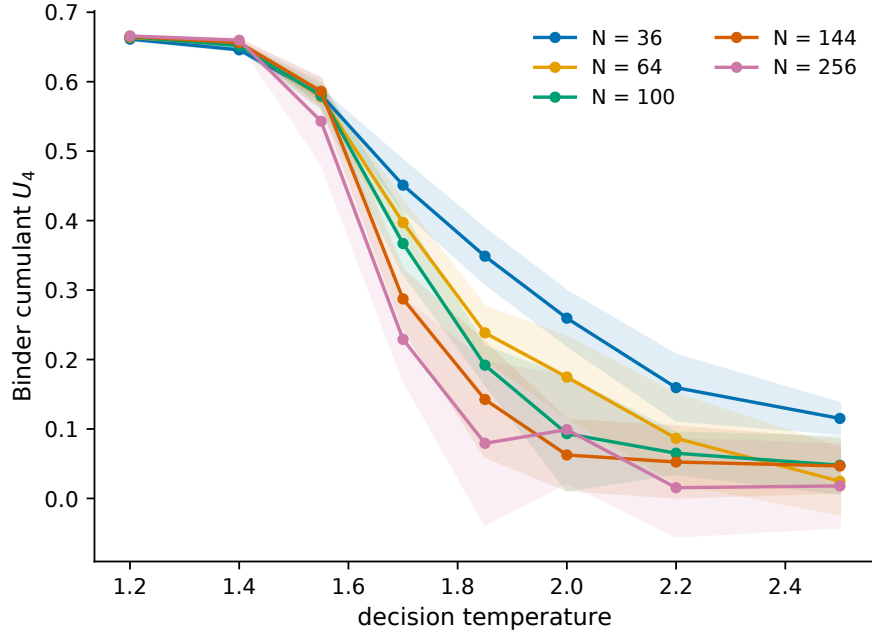


Figure 5: Binder cumulants across five finite sizes. Shading gives 95% normal intervals across six independent realizations.

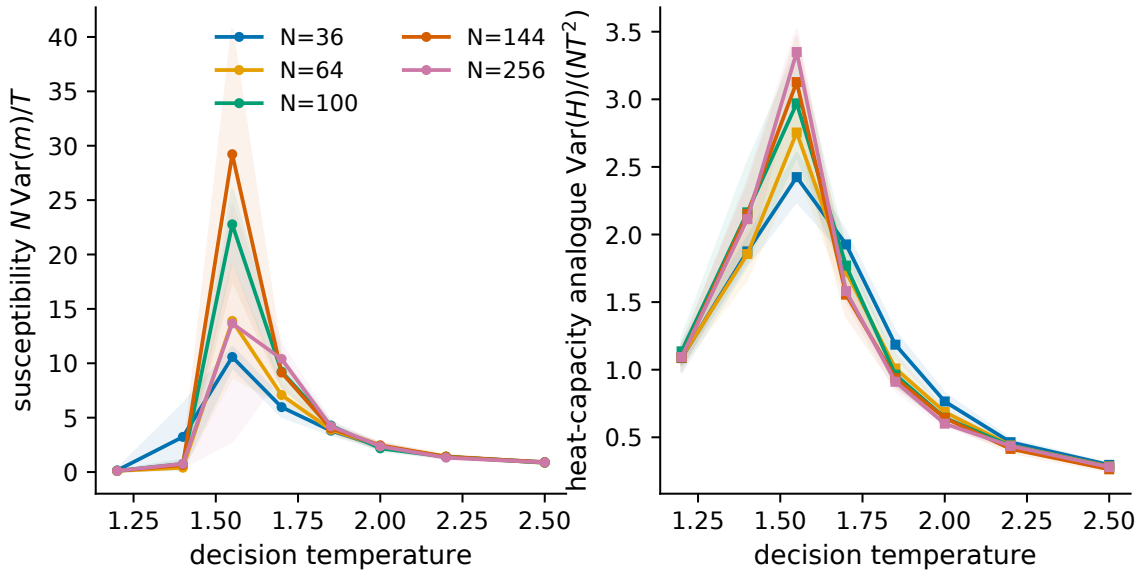


Figure 6: Susceptibility and heat-capacity analogue across the frozen temperature grid.

8 Metastability, hysteresis, and relaxation

The exact constrained free-energy profile and a finite-rate field sweep are shown in figure 7. A loop is evidence of finite-time metastability, not by itself of equilibrium coexistence. Relaxation from an ordered state in figure 8 is long or censored below the numerical boundary and drops sharply above it. This establishes slow finite-time persistence, but the coarse temperature grid and censoring rule do not isolate an asymptotic critical-slowing-down exponent.

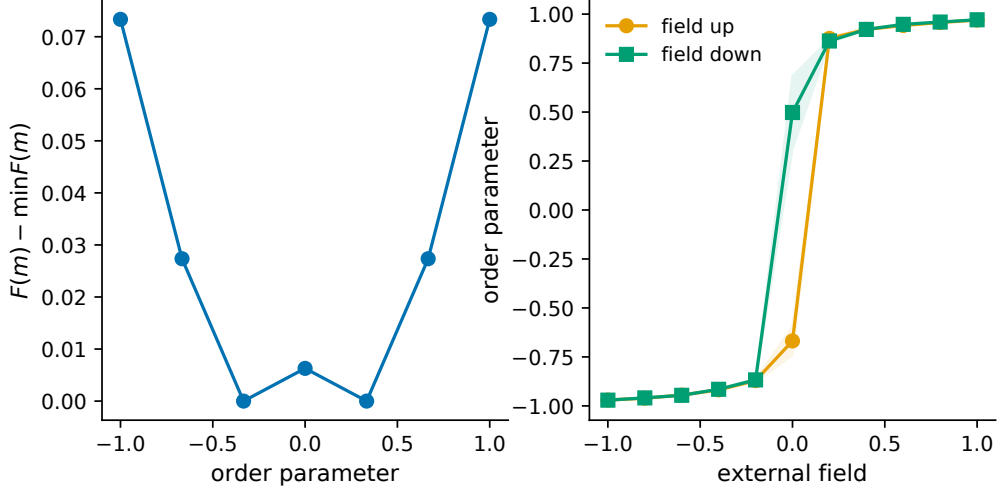


Figure 7: Exact small-system constrained free energy (left) and numerical finite-rate field cycling (right).

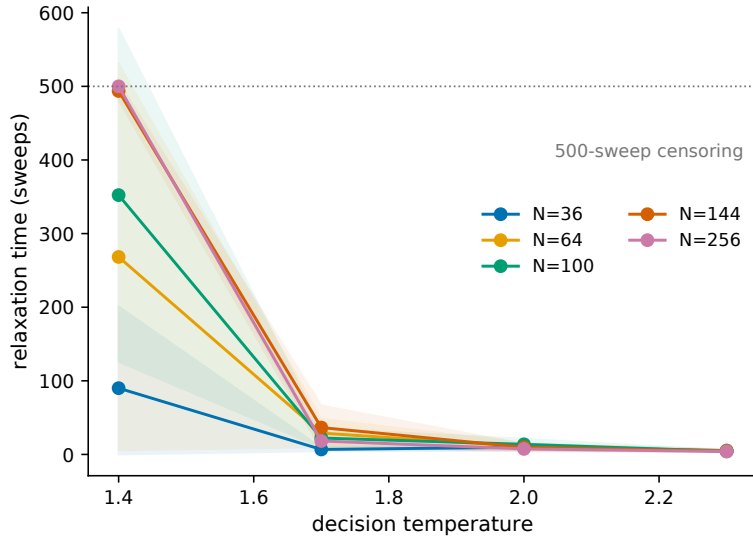


Figure 8: Relaxation time from the ordered state. The operational threshold is fixed at $|m| \leq e^{-1}$; trajectories that do not cross are censored at the frozen horizon.

9 Application mappings

The abstract model is mapped, without parameter retuning, to humanitarian multi-commodity coordination and defensive cyber-utility restoration. The humanitarian drive injects correlated workload, constrains shared service capacity, and allows unresolved shortage to propagate on the dependency layer. The utility drive removes service, corrupts selected local fields, makes communication asymmetric, imposes a temporary partition, constrains crews, and allows dependency-mediated secondary failures. No offensive cyber procedure or real infrastructure target is represented.

Figure 9 shows one prespecified snapshot from each mapping. Figure 10 shows the utility ensemble trajectory. Workload and resources are reconstructed from separate injected, cascaded, completed, and remaining terms; the maximum residual is 4.44×10^{-14} .

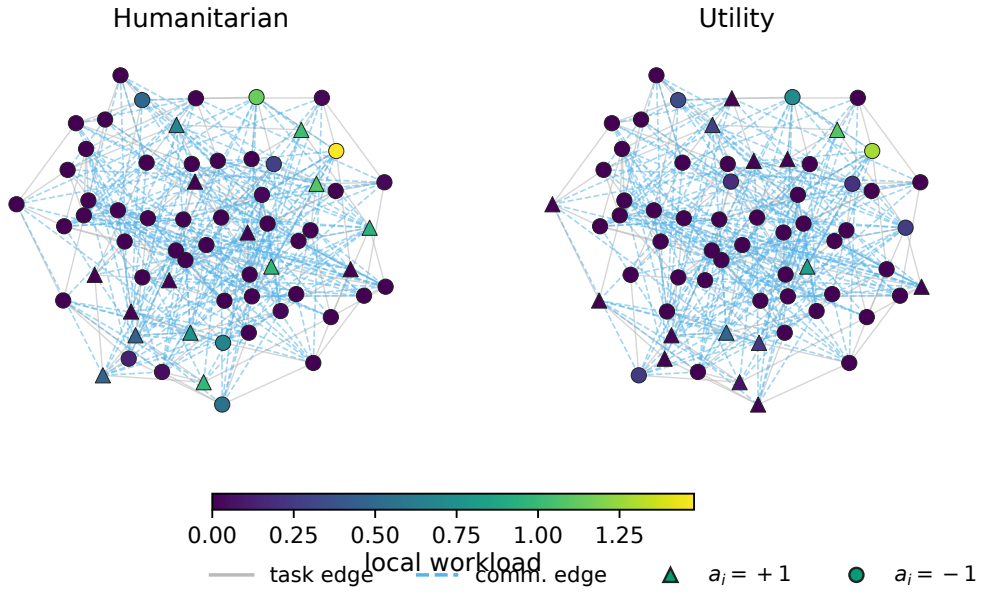


Figure 9: Humanitarian and utility multiplex snapshots during the driven interval. Node color is local workload; shape is the independently selected action. Solid gray and dashed blue edges denote dependency and communication.

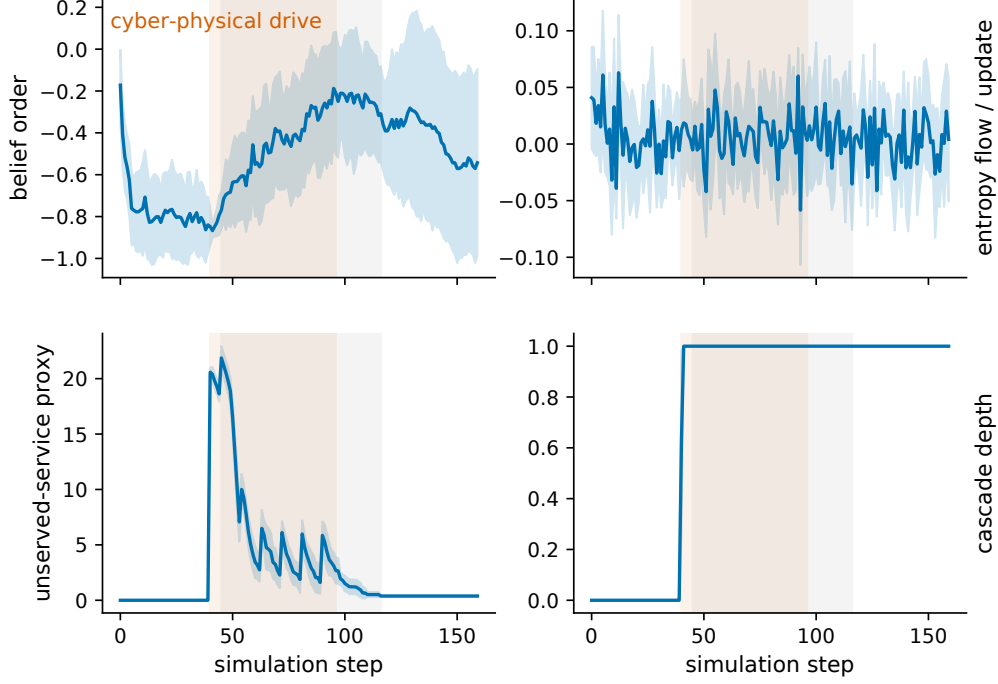


Figure 10: Defensive utility mapping under an abstract cyber-physical drive and communication partition. Bands are 95% intervals across independent trajectories. Transition entropy flow is a logged affinity statistic, not a separately proven fluctuation-theorem entropy production.

10 Discussion

The principal conceptual result is a boundary between a literal mathematical construction and an analogy. Equation (2), its Gibbs law, detailed balance, and equation (9) are exact in the stated limit. Decision temperature is a policy-noise parameter. Once communication is directed or delayed, memory and workload evolve, or the environment is driven, the equilibrium Hamiltonian no longer governs the full process. Entropy production then follows from transition currents, not from an operational severity score.

The numerical results support an exact equilibrium reference, positive current-based entropy production under directed coupling, and a topology- and disorder-dependent finite-size crossover. They do not prove a thermodynamic-limit singularity, universal critical exponents, literal physical heat, or field validity of either application mapping. The generalized-entropy comparison gives correlations of 0.951–0.986 with Shannon across the frozen cells and no independent predictive or scaling advantage; consequently Shannon entropy remains the default unless a future model establishes nonextensive structure independently.

Autonomy matters to the interpretation. Each local update belongs to a persistent agent and can change under a private-observation counterfactual without mutating peer state. Nevertheless, the binary model is not a language model, a trained reinforcement-learning policy, or a high-fidelity logistics or power-system solver. Its value is analytical transparency.

11 Conclusion

We have given a decentralized belief–action system an explicit microstate, Hamiltonian, reversible local policy, Gibbs reference law, and controlled nonequilibrium extensions. Exact enumeration verifies the equilibrium identities; directed interaction yields positive probability-current entropy production; and finite systems exhibit a topology-, communication-, and disorder-dependent coordination crossover with response and relaxation signatures. The result is a statistical-mechanics formulation of agentic coordination, not evidence that entropy is a superior controller.

A Exact state enumeration

For N agents the binary state space has 2^{2N} configurations. We enumerate all states, calculate $H(x)$, normalize equation (7) with a shifted log weight, and construct the full asynchronous heat-bath kernel. The stationary distribution of a directed kernel is obtained by solving $P^\top \boldsymbol{\pi} = \boldsymbol{\pi}$ with normalization. All matrix rows and probability vectors are checked for finite values and unit mass.

B Reproduction and provenance

The immutable parent is V8 commit `b86f97fa0940f11cb366c809e0e46fa888dfaba1`. V9 is intentionally left uncommitted for review. The frozen protocol and source checksums are recorded in full in the accompanying reproducibility summary (prefixes `8520d841f987` and `9845f4d6ef2c`, respectively). Exact commands, external raw-artifact hashes, compact tables, and figure source data accompany the repository. No human participants, LLM inference, or GPU training are part of this study.

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